

GS SUSTAIN INSIGHTS

Demographics to Demand: Takeaways from call on drivers, myths and impacts of aging & declining populations

On July 21 we hosted a webcast with Dr. Dean Spears — Economic Demographer, Development Economist, Associate Professor of Economics at the University of Texas at Austin, and author of ‘After the Spike: Population, Progress and the Case for People’. We discussed the history of populations rise and its forward trajectory, and explored the economic, environmental, and social implications of aging and declining populations – covering what our presenter sees as misperceptions around the benefits and downsides of population decline. We summarize key takeaways below:

1. As fast as populations rose they may fall — Humanity currently past peak births and approaching peak populations: Our presenter highlighted that for thousands of years human numbers grew slowly with only ~300 million people at the start of the common era, but improvements in survival — better nutrition, sanitation, and reduction in child mortality drove exponential growth, reaching 1 billion by the early 1800s, 2 billion by 1900s and quadrupling over the last century. Critically, birth rates have actually been falling over the long arc of history even as population grew – 1968 was the year of fastest population growth and it has never been that fast since. According to our presenter, humanity reached ‘peak births’ in 2012, but not yet ‘peak human’ because lifespans are still extending. Any sustained global birth rate below replacement roughly 2 children per 2 adults results in long-term depopulation. At a total fertility rate (TFR) of 1.5 (between Europe and US levels), each successive generation shrinks meaningfully, falling by nearly half by the third generation.

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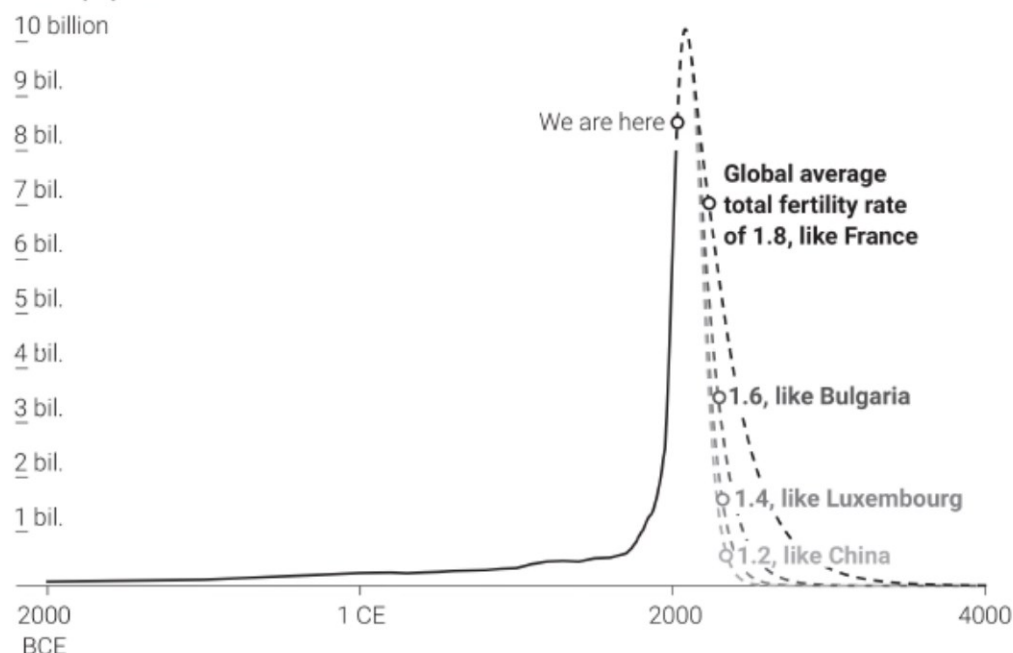
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Exhibit 1: Any long-term birth rate that averages under two would cause depopulation

World population in billions under different TFR scenarios

World population

Source: Spears, et al. (2024)

2. Peak populations coming in earlier than expected in both developed and developing countries, with future estimates heavily dependent on fertility and rebound assumptions.

According to our presenter, the UN projects the global population peak in 2084, but this rests on an assumption that low-fertility countries (Japan, China, much of Europe and Latin America) will rebound back toward ~1.6, based on models fitted to early-2000s European data that proved non-durable. Our presenter remains skeptical of the fertility rebound, finding that every US state's fertility rate was lower in the latest data than in 2010, and every Japanese prefecture is lower than a decade ago – suggesting the peak will come sooner than the UN's 2080s estimate. The UN's own low fertility projections point to global population peaks in the 2050s, aligning more closely with many demographers' views based on our conversations. The UN's base estimates indicate developed economies have already reached peak population.

Importantly, low fertility is no longer a rich-country phenomenon as two-thirds of people now live in countries with birth rates below 2, including Mexico (below the US), India (below replacement), and Latin America as a whole at ~1.8. According to our presenter, even in disadvantaged regions like Uttar Pradesh in India, where many young women lack mobile phones or internet, women report wanting fewer than 2 children, undercutting single-cause explanations. Sub-Saharan Africa remains the key question mark at TFR of 4.3, but its declining rates track expected levels for its human development stage and are likely to keep falling according to our presenter. *For more detail on latest trends in falling fertility see **DM population decline & rapidly falling birth rates amid the rise of AI**.*

3. Drivers of falling fertility are broad-based with no single explanation holding and no 'shovel ready' policy solutions. According to our presenter, an 'all-of-the-above' answer is likely the best starting point for explaining falling fertility rates as the decline is too large and globally shared for any single explanation (smartphones, religion, delayed

marriage / coupling rates, feminism) to suffice. Even religious and marriage-heavy societies like India and Latin America show low rates, as do both US red and blue states, according to our presenter. In our presenter's view, the unifying thread of opportunity costs of having children is rising as the world offers more opportunities with education, careers, and even leisure/dopamine trade-offs like video games and streaming. There remains no 'shovel ready' solutions to solving declining fertility rates, according to our presenter, as many pro-natalist policies like baby bonuses have shown limited success. However, it still remains early days for learning and our presenter argues that creating a more equal and fairer parenting society could help and raising fertility back to replacement levels wouldn't require sacrificing equality progress, as some OECD countries with the largest women-wage gap show some of the lowest fertility rates.

4. Myths on Environmental and Social Benefits of Depopulation. Contrary to many beliefs around benefits of smaller populations, **humans have continued to improve environmental outcomes despite growing populations** — our presenter points to examples of China's air pollution in 2013 has been reduced dramatically despite population increases, emissions per person has dramatically declined, etc. Additionally, our presenter emphasized that decarbonization requires action today, whereas demographic change unfolds over decades. Declining births now offer little benefit for reducing emissions by 2050, since children not born today would not become major consumers for another 20-plus years. In our view, broader debates will mount over how to prioritize allocation of government resources in an aging society. This could impact support for environmental initiatives—especially where benefits are not perceived as immediate—as fiscal support for social programs like retirement and healthcare will likely take priority. **Socially**, other people should be viewed as win-win through the broader exchange of ideas and goods rather than competition, and avoiding depopulation does not require rolling back gender progress according to our presenter. More people = more ideas, more creativity, more data and support for investment (especially new drug discoveries) that benefits all humanity.

5. Economic and Innovation Implications of Depopulation: Unit fixed costs rise and economic options erode. Our presenter argues the core economic benefit of scale effects from fixed and marginal costs (also called preference externalities) will be eroded - where large populations help cover fixed costs allowing for more variety, customization, lower prices, better infrastructure. **Larger populations help cover fixed costs** — from drug discoveries to producing movies, our presenter sees a declining population over time potentially driving less infrastructure expansion. Drug companies would reallocate R&D toward larger age cohorts, movies wouldn't get produced because of a shrinking audience, businesses would operate fewer days, and there could be destructive competition over a shrinking pie. In our view, where markets or volume are in decline, companies will need to consider product redesign / reformulation targeting an aging demographic, international expansion to remedy declining domestic demographics, and launch new service offerings to grow and maintain customer value over the long term. On the topic of AI, our presenter sees AI complementing humans (like Excel or pencils historically), though fewer people would lead to fewer productive human-AI pairs. Finally, fiscal pressures (pensions, healthcare) would force some combination of benefit cuts, later retirement, or higher taxes, in his view.

GS SUSTAIN investment view of the economic and company demand implications of

aging & declining populations

In our recent installment of our Demographic Dilemma series, we evaluated the growing influence of spending shifts in aging populations and how they will impact product, industry, and company demand. **We introduced our Demographic-Driven Demand (DDD) framework — covering ~3,000 companies** (*available upon request*) — to identify which industries/companies have the greatest demographic-driven demand impact vs. a 2023 baseline before accounting for product differentiation or shifts in corporate strategy.

While the demographic shifts of aging, rapidly falling fertility rates, and declining young populations — developed economies are set to experience retiree growth (ages 65+) of 4 million annually, while core consumers aged 35-55 are set to see accelerated declines of 3 million p.a. starting in 2030 (UN) — have long been topical, we see potential for a material impact by the end of the decade that could drive headwinds/tailwinds for product demand and influence corporate strategies around regional mix, product mix and integration/consolidation.

Industries with Demographic Headwinds:

- **Apparel and Footwear** face significant headwinds associated with declining youth populations, some of the largest historical spenders on new apparel.
- **Autos are at a strategic crossroads** amid declining prime new driving age populations, and declining spending among older age cohorts.
- **Other notable industries facing headwinds:** Education, Electronics (PCs, tablets, TVs), Food Away From Home, Airports, Public Transportation (vehicle / rail).

Industries with Demographic Tailwinds:

- **Healthcare**, in our view, will see the strongest demographic demand from aging. **However, Healthcare and Medtech are not immune to demographic shifts as growth in older populations is set to decelerate from prior decades.**
- **Leisure and Entertainment:** Home Improvement & Furnishings, Reading, Television watching, Cruise Lines, Discount Retailers, and Pets.
- **Residential Electricity Consumption:** Spending data shows older age cohorts spend more on utilities and use more electricity, likely given excess time at home.
- **Senior Living & Aging at Home:** 70% of individuals over 65 prefer to age in their home, presenting opportunities for home healthcare, telemedicine and digital devices.
- **Other notable industries facing tailwinds:** Include Home Improvement & Furnishings, Reading, Retail – Staples and Multiline.

Select themes and Buy-rated companies with demographic tailwinds highlighted by our analysts (not inclusive of all demographic beneficiaries) — **Senior Living:** Ventas; **Home Improvement:** Home Depot, Lowe's, Sherwin-Williams; **Leisure & Entertainment:** Viking Holdings, Carnival Corp, Royal Caribbean, H World Group (China Hospitality); **Discount Retailers:** Ross Stores, TJX; **MedTech:** Smith & Nephew (on CL); **Healthcare:** HCA Healthcare, UnitedHealth, Alignment Healthcare, Rede D'Or (Brazil hospitals), Daiichi Sankyo, and Sysmex.

For more information, please read - [GS SUSTAIN: Demographics to Demand: How Aging and Declining Populations Will Drive Product and Company Demand Tailwinds/Headwinds.](#)

Exhibit 2: Select Companies exposed to Aging Demographics and Spending Shifts

Key Aging Demographic Beneficiaries and Stocks Exposed

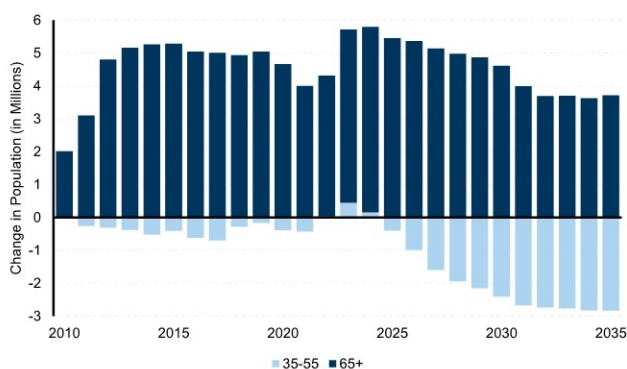


Source: Goldman Sachs Global Investment Research

Demographics to Demand - Summary in charts

Exhibit 3: In developed economies, retirees are growing ~4 mn pa, while peak consumers (35-55) are set to decline ~3 mn a year over the next decade

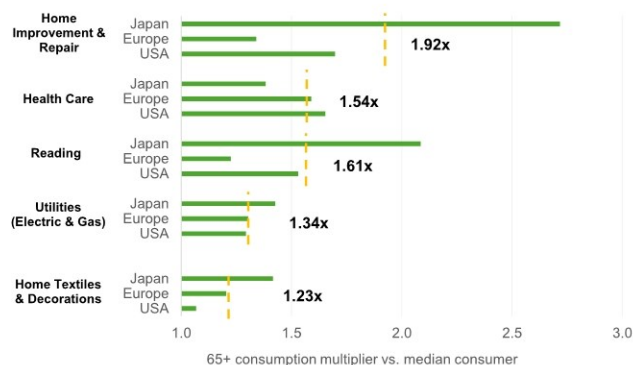
Annual change in population (millions) by age cohort in More Developed Regions, 2010-2035E; medium variant estimate 2024 onwards



Source: UN Population Prospect, Data compiled by Goldman Sachs Global Investment Research

Exhibit 4: 65+ spend more on Home Improvement, Healthcare, Reading, Utilities, and Decorations...

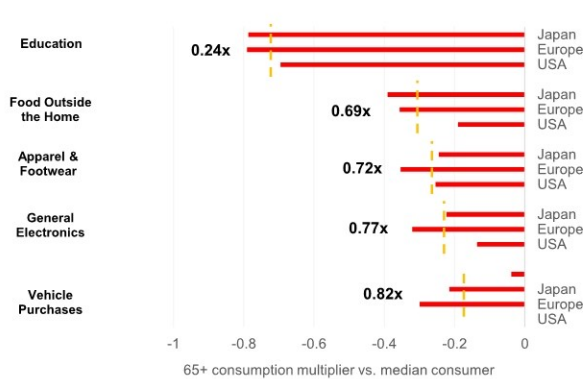
Top 5 categories with highest 65+ consumption multiplier vs. median consumer by region



Source: BLS, Eurostat, SBJ, Data compiled by Goldman Sachs Global Investment Research

Exhibit 5: ..while spending less on Education, Food Outside the Home, Apparel & Footwear, Electronics, and Vehicle Purchases

Bot. 5 categories with lowest 65+ consumption multiplier vs. median consumer by region



Source: BLS, Eurostat, SBJ, Data compiled by Goldman Sachs Global Investment Research

Exhibit 7: Our DDD framework can provide insights into company's demographic headwinds and tailwinds

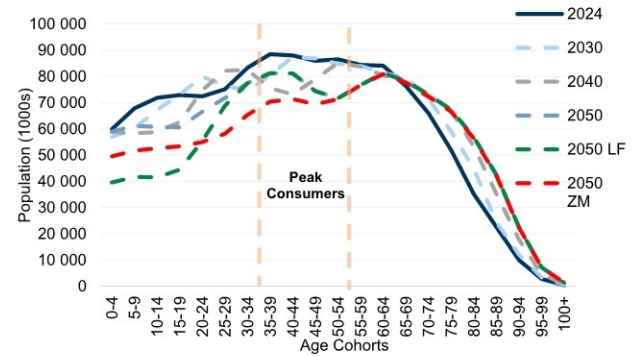
Headline and country level demographic-driven demand growth, 2030-2050 vs 2023

Nike Inc				
Demographic-Driven Demand				
		2030	2040	2050
UN Base Case		0.9%	1.6%	1.5%
Country Level Demographic-Driven Demand Impact				
Country	% of Revenue	2030	2040	2050
United States of America	36%	2.6%	6.0%	8.5%
China	13%	-3.7%	-10.3%	-17.5%
Canada	6%	5.0%	9.7%	13.1%
Japan	2%	-4.6%	-11.2%	-16.7%
India	2%	5.4%	10.8%	12.7%
Others	22%	0.8%	0.9%	0.6%

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 6: The developed world has likely reached peak consumer population aged 35-55, and will be significantly lower under zero migration / low fertility scenarios in the future

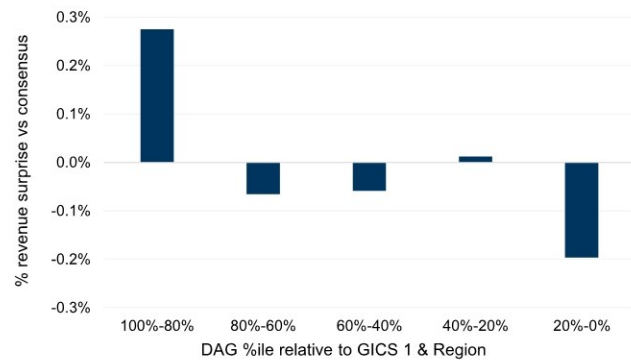
Population distribution by age cohorts 0-100+, 2024-2050 medium variant



Source: UN Population Prospects, Data compiled by Goldman Sachs Global Investment Research

Exhibit 8: Companies with better demographic outlooks have seen higher average revenue surprises over the past three years

Trimmed mean (80%) % revenue surprise vs. consensus estimates by DDD quintile (GICS 1 & Region relative), 2023-2025



Source: FactSet, Goldman Sachs Global Investment Research

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